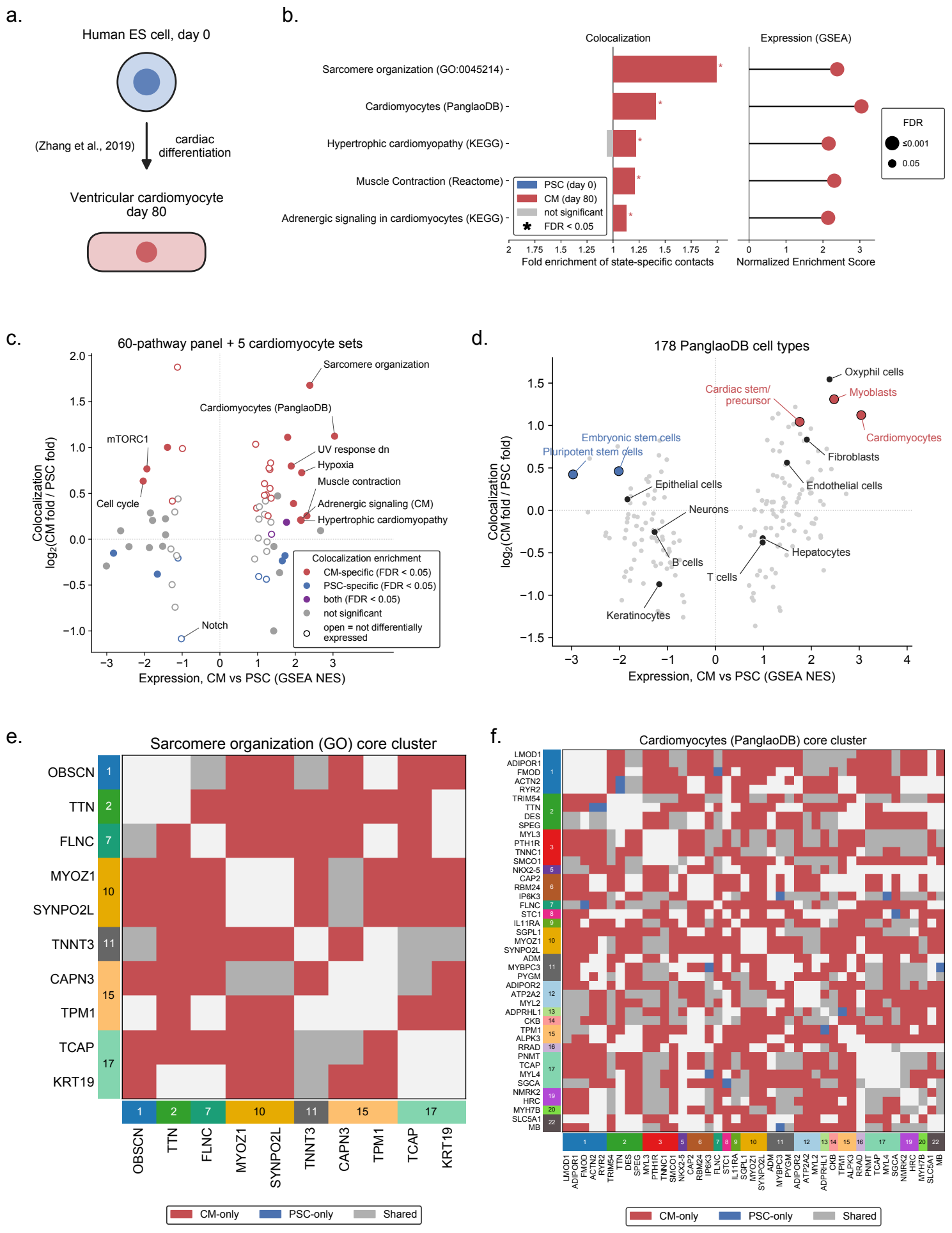

a.
Human ES cell, day 0
(Zhang et al., 2019)
cardiac differentiation
Ventricular cardiomyocyte day 80
b.
Colocalization
Expression (GSEA)
Sarcomere organization (GO:0045214)
Cardiomyocytes (PanglaoDB)
Hypertrophic cardiomyopathy (KEGG)
Muscle Contraction (Reactome)
Adrenergic signaling in cardiomyocytes (KEGG)
PSC (day 0)
CM (day 80)
not significant
FDR < 0.05
FDR
≤0.001
0.05
Fold enrichment of state-specific contacts
Normalized Enrichment Score
c.
60-pathway panel + 5 cardiomyocyte sets
Sarcomere organization
Cardiomyocytes (PanglaoDB)
mTORC1
UV response dn
Hypoxia
Muscle contraction
Cell cycle
Adrenergic signaling (CM)
Hypertrophic cardiomyopathy
Notch
Colocalization enrichment
CM-specific (FDR < 0.05)
PSC-specific (FDR < 0.05)
both (FDR < 0.05)
not significant
open = not differentially expressed
Colocalization log2(CM fold / PSC fold)
Expression, CM vs PSC (GSEA NES)
d.
178 PanglaoDB cell types
Oxyphil cells
Myoblasts
Cardiac stem/ precursor
Cardiomyocytes
Embryonic stem cells
Pluripotent stem cells
Fibroblasts
Epithelial cells
Endothelial cells
Neurons
Hepatocytes
B cells
T cells
Keratinocytes
Colocalization log2(CM fold / PSC fold)
Expression, CM vs PSC (GSEA NES)
e.
Sarcomere organization (GO) core cluster
OBSCN
TTN
FLNC
MYOZ1
SYNPO2L
TNNT3
CAPN3
TPM1
TCAP
KRT19
CM-only
PSC-only
Shared
f.
Cardiomyocytes (PanglaoDB) core cluster
CM-only
PSC-only
Shared